

A Complete Optoelectronic and Thermodynamic Characterization for $\text{Mo}_{1-x}\text{W}_x\text{Se}_2$ Alloy: a Data-Driven Workflow Approach

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Motivation

Engineering 2D Materials

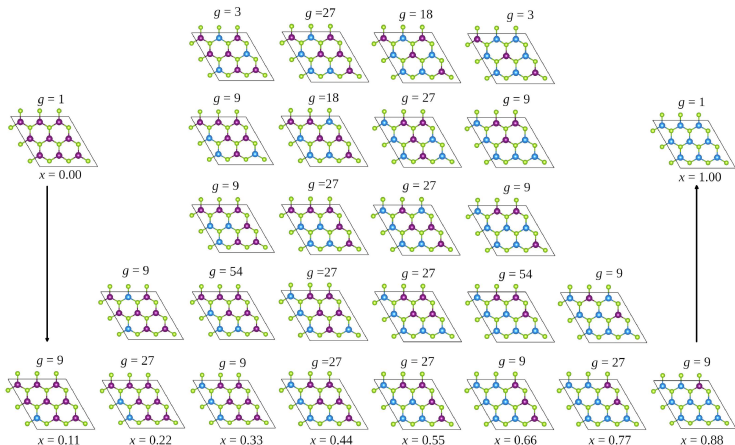


- ▶ Transition Metal Dichalcogenides (TMDs) are a promising class of 2D materials for next-generation optoelectronics.
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- ▶ **Key Challenges:**
 - ▶ Understanding the thermodynamic stability.
 - ▶ Accurately predicting optical properties, which are dominated by strongly bound electron-hole pairs (excitons).
 - ▶ These simulations are computationally demanding. For example, alloys with 0.1% of Mo is necessary a supercell $32 \times 32 \times 1$, i.e, 3072 atoms.

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 - ▶ These simulations are computationally demanding. For example, alloys with 0.1% of Mo is necessary a supercell $32 \times 32 \times 1$, i.e, 3072 atoms.
- ▶ **Our Goal:** To develop an efficient protocol to investigate the thermodynamic and optoelectronic properties of TMD alloys with excitonic effects.

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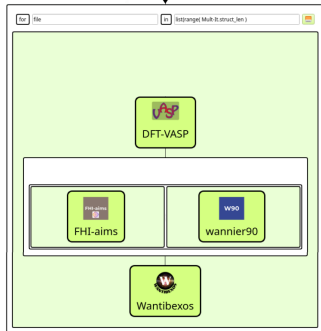
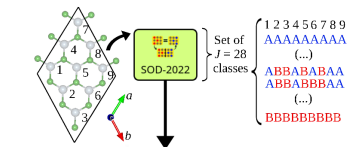
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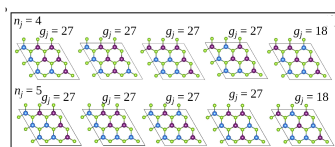
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5. **Statistical Averaging (GQCA):** Compiled all results to compute thermodynamic and optoelectronic properties as a function of composition (x) and temperature (T).

Our Approach

Key Steps in the Workflow



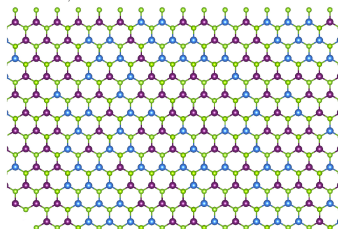
colab
Statistical ensemble



Generalized ensemble: random alloy behaviour

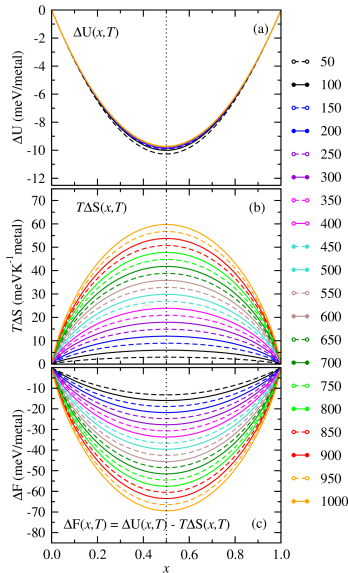
$$\Delta S = -Nk_B [x \ln x + (1-x) \ln(1-x)] \left\{ \begin{array}{l} \text{based on Kullback-Leibler divergence} \end{array} \right.$$

$$\Delta S = Nk_B \sum_{j=1}^J [\ln(g_j) - \ln(x_j)] \left\{ \begin{array}{l} \text{Shannon entropy} \end{array} \right.$$



Results

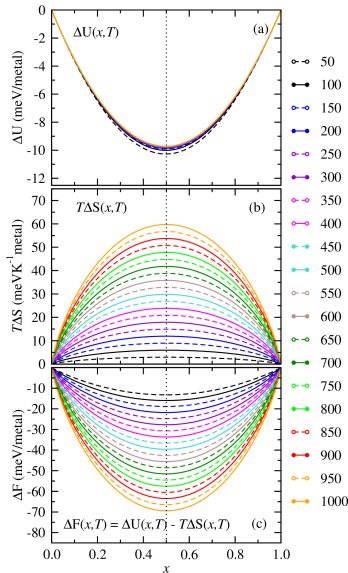
Thermodynamic Stability



- The internal energy (ΔU) is negative.
- The Helmholtz free energy ($\Delta F = \Delta U - T\Delta S$) is negative over a broad temperature range.
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Main Finding

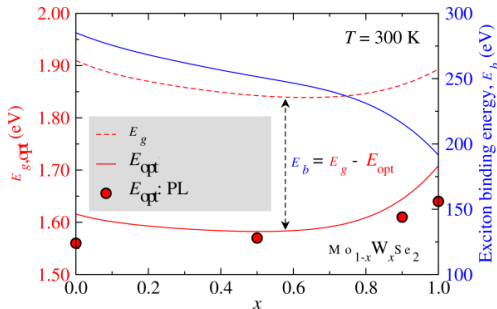
$\text{Mo}_{1-x}\text{W}_x\text{Se}_2$ is a **stable random alloy** across all compositions and temperatures investigated.

Results

Electronic vs. Optical Band Gaps

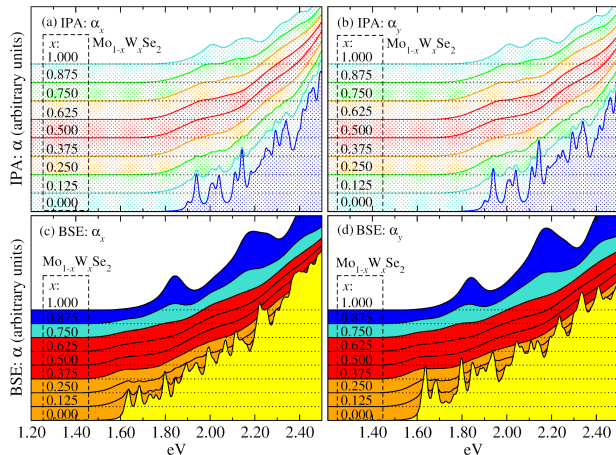


- ▶ Gaps exhibit a **non-linear dependence** on composition (bowing effect).
- ▶ For W-rich compositions, the gaps increase significantly due to the stronger **Spin-Orbit Coupling (SOC)** effect of W.
- ▶ E_b is large but **decreases** as W concentration increases.
- ▶ This reduction is due to strong SOC in W-rich alloys, which modifies the electronic band curvatures.



Results

Optical Absorption Spectra



- ▶ Excitonic effects are essential for accurately predicting optical properties.
- ▶ The BSE spectra show a **red-shift** for Mo-rich compositions (due to higher E_b).
- ▶ For W-rich compositions, SOC effects become dominant, leading to a reduction in absorption intensity.

- ▶ We successfully developed a cost-effective workflow to provide a complete thermodynamic and optoelectronic characterization of $\text{Mo}_{1-x}\text{W}_x\text{Se}_2$ alloys.
- ▶ Our statistical model confirms that $\text{Mo}_{1-x}\text{W}_x\text{Se}_2$ forms a **stable random alloy** over a wide range of temperatures and compositions.
- ▶ The optoelectronic properties show a strong **non-linear dependence** on composition, governed by the interplay between excitonic effects and Spin-Orbit Coupling.
- ▶ This workflow provides a robust tool for the rational design of novel TMD alloys for future optoelectronic devices.

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First-Principles Statistical Investigation of Thermodynamic Behavior with Excitonic Effects in $\text{Mo}_{1-x}\text{W}_x\text{Se}_2$ Alloys through a Data-Driven Workflow Approach[†]

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We investigate thermodynamic stability of the $\text{Mo}_{1-x}\text{W}_x\text{Se}_2$ alloy at the atomistic level and introduce an innovative cost-effective protocol for predicting and characterizing its optoelectronic properties, explicitly

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 - ▶ Brazilian Synchrotron Light Laboratory (LNLS)

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